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Communication Systems II

Seeing Signals

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Abstract

The present study investigates the problem of Automatic Modulation Classification (AMC) and the characterization of physical-layer impairments in modern telecommunication systems, utilizing Deep Learning methods. For the purposes of this research, an extensive and realistic dataset (76, 800 samples) was prepared via simulation, which includes 16 different modulation schemes (ASK, PSK, QAM, HQAM, APSK) under the influence of additive noise (AWGN), phase noise (Phase Noise), I/Q imbalance (I/Q Imbalance), and external interference (Interference). The signals were converted into two-dimensional histograms (Cartesian and Polar images) and analyzed through two radically different approaches: a specialized Multi-Task Convolutional Neural Network (Multi-Task CNN) and a state-of-the-art, generative Multimodal Model (`SmolVLM-256M-Instruct`) adapted via the LoRA technique.

The central axis of the evaluation was the comparative analysis of the two architectures in terms of classification accuracy, robustness in low-SNR environments, sensitivity in detecting distortions, and generalization capability in intermediate, unseen channel conditions. The results demonstrate the quantitative superiority and dramatically lower computational complexity of the CNN, making it ideal for real-time applications. Simultaneously, however, the impressive capability of the VLM to achieve comparable accuracy rates using a minimal fraction of the training data (sample efficiency) is proven, highlighting the potential of Large Foundation Models.

1. Literature Review

In modern wireless networks, automatic modulation classification (AMC) and the detection of physical-layer impairments constitute fundamental functions for dynamic spectrum allocation [2]. In recent years, Convolutional Neural Networks (CNNs) have been the established solution, utilizing constellation diagrams as images for signal classification. However, conventional CNNs remain single-task and often lag in generalization capability when multiple, complex distortions coexist in the signal.

Recently, Multimodal Vision-Language Models (VLMs), such as the RF-GPT architecture [1], were introduced into telecommunications. These models treat signal analysis as a Visual Question Answering problem. The present study evaluates this new trend, comparing a multi-parameter (Multi-Task) CNN with a lightweight VLM (`SmolVLM-256M-Instruct`) trained via LoRA, with the goal of testing their generalization capability under unseen noise conditions (unseen conditions).

2. System Model and Dataset Construction

2.1 Signal Generation and Energy Normalization

Each digital symbol $x[n] = I[n] + jQ[n]$ belongs to one of the 16 constellations defined in the literature [2] (ASK, PSK, QAM, HQAM, APSK). Particular emphasis was placed on their accurate geometric representation, such as the design of hexagonal lattices for HQAM schemes and concentric rings with phase offset for APSK. To ensure a fair comparative analysis, all generated symbols were normalized in energy so that their average power strictly equals unity ($E_s = 1$).

2.2 Transceiver Impairment Modeling (Impairments)

Realistic distortions were applied to the ideal signal through targeted mathematical models:

i. **AWGN:** Additive white Gaussian noise, controlled via the SNR (0–30 dB), which was categorized into three severity levels (*Severe*: 0–10 dB, *Medium*: 10–20 dB, *Low*: 20–30 dB).

ii. **Phase Noise:** Modeled via a wrapped Gaussian distribution [2], causing random phase drift and rotation of the constellation.

iii. **I/Q Imbalance:** Introduction of gain and phase asymmetry between the I and Q branches, distorting the orthogonality of the system.

iv. **Interference (Jamming) & Distortion:** Addition of a continuous wave (CW) interferer of random frequency, as well as simulation of non-linear amplitude distortion.

2.3 Conversion to Images (Cartesian/Polar) and Structured Labeling

For their compatibility with computer vision networks, the distorted complex sequences $y[n]$ are converted via a two-dimensional histogram into two types of images (grayscale, 256×256 pixels) that capture the probability density of the symbols:

i. **IQ Images (Cartesian):** Based on the components $I = \mathrm{Re}\{y[n]\}$ and $Q = \mathrm{Im}\{y[n]\}$, producing the classic constellation diagram.

ii. **Polar Images (Polar):** Based on the magnitude and phase of the symbols, namely the coordinates $(|y[n]|, \angle y[n])$, offering an alternative geometric representation of the distortions.

During bulk generation, each pair of images was associated with a common vector of structured labels (ground truth labels) that answers the following classification questions: modulation type, SNR range, and presence or absence (`True`/`False`) of phase noise, I/Q imbalance, and external interference. The images below along with their labels constitute random samples of the set of images that were produced.



Figure 1: 2D histograms of received symbols (grayscale) for the 8-ASK modulation scheme: `sample_id: 5743`, `severity_level: Low`, `snr_db: 26.86`, `phase_noise_present: Yes`, `iq_imbalance_present: No`, `external_interference: Yes`.



Figure 2: 2D histograms of received symbols (grayscale) for the 64-HQAM modulation scheme: `sample_id: 55457`, `severity_level: Medium`, `snr_db: 16.18`, `phase_noise_present: Yes`, `iq_imbalance_present: Yes`, `external_interference: Yes`.

2.4 Theoretical Performance Evaluation (Baseline SEP)

Prior to the training of the CNN and VLM models, a Monte Carlo simulation was conducted to extract the Symbol Error Probability (SEP) as a function of SNR, setting a strict theoretical benchmark for traditional receivers. The analysis was performed on standard square M-QAM modulations ($M \in \{4, 16, 64, 256\}$) and Cross-QAM ($M \in \{32, 128\}$). Each transmitted symbol $x \in \mathcal{S}$, where $\mathcal{S}$ is the ideal constellation, underwent distortion according to the channel model:

$$y = x \cdot e^{j\phi} + w$$

where ϕ is the phase noise, which follows a wrapped Gaussian distribution with standard deviation $\sigma_\phi = 0.05$, and w is the additive white noise (AWGN).

At the receiver, the optimal Maximum Likelihood detector was implemented, which makes decisions by minimizing the Euclidean distance between the received signal y and all possible ideal symbols of the constellation:

$$\hat{x} = \arg\min_{s \in \mathcal{S}} |y - s|$$

The generation of the SEP curves (Figure 3) was implemented for an SNR range from 0 to 30 dB with a step of 2 dB, studying $50,000$ independent symbols per noise level, confirming the dramatic performance collapse in higher-order modulations due to their high sensitivity to phase fluctuations.

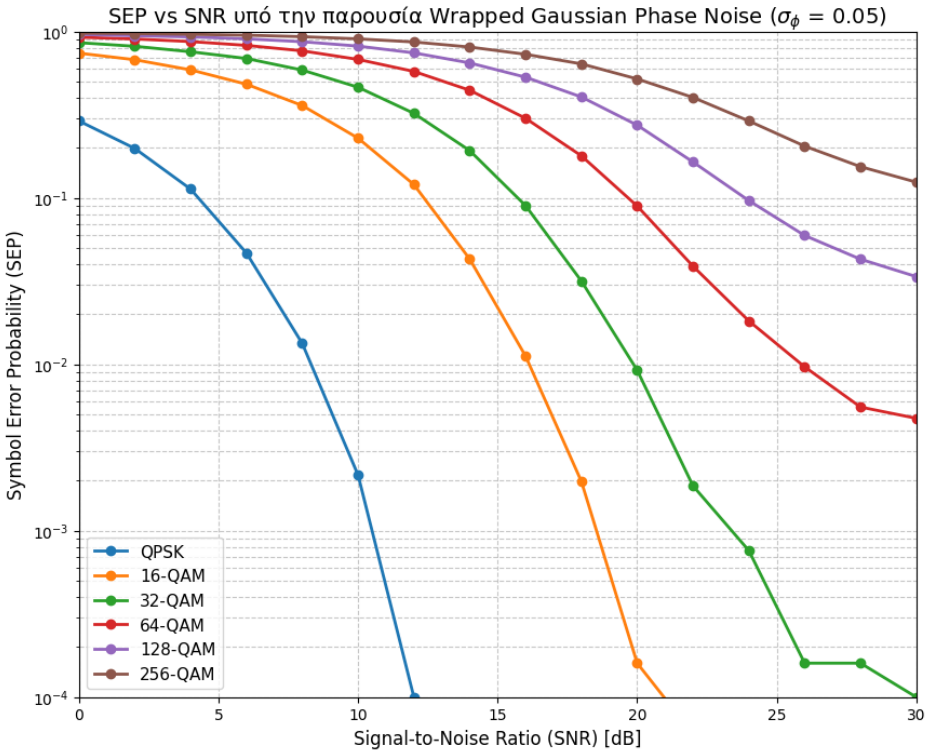


Figure 3: Symbol Error Probability (SEP) curves as a function of SNR for a traditional minimum distance receiver. The simulation includes M-QAM modulations under the presence of AWGN and phase noise ($\sigma_\phi = 0.05$).

3. Machine Learning Model Architectures and Training

The goal of the present chapter is the development and evaluation of deep learning models for Automatic Modulation Recognition (AMR) and telecommunication channel characterization. It is worth noting that the entirety of the experimental procedure, including the generation of the visual datasets (I/Q histograms) and the training of both architectures (CNN and VLM), was implemented exclusively in the cloud via the Google Colab environment. To meet the massive requirements in memory and processing power, an NVIDIA A100 hardware accelerator (GPU) was utilized exclusively.

3.1 Multi-Task CNN Architecture

The proposed model was designed to simultaneously extract multiple pieces of information from the two-dimensional histograms (images) of the received signals. The architecture implements the philosophy of Multi-Task Learning (MTL) [3] and is divided into two main parts:

i. **Feature Extraction Backbone (Backbone):** Consists of three sequential convolutional layers (`Conv2D`) with 16, 32, and 64 filters respectively. Each convolution is followed by a ReLU activation function and subsampling layers (Max-Pooling). The final stage of the backbone applies adaptive average subsampling (Adaptive Average Pooling) to condense spatial information prior to flattening (Flatten).

ii. **Independent Classification Heads (Classification Heads):** The extracted features are fed into five parallel, fully connected sub-networks, each tasked with a separate job:

- Two multi-class classification heads for predicting the modulation type (16 classes) and the SNR degradation level (3 classes).

- Three binary detection heads, using a Sigmoid activation function at the output layer, for detecting the presence of phase noise, I/Q imbalance, and external interference respectively.

3.2 Training Strategy and Data Splitting

To ensure academic validity and avoid overfitting, the dataset of 76,800 samples was strictly divided into three subsets: Training (**Train Set** — 70%), Validation (**Validation Set** — 15%), and Testing (**Test Set** — 15%).

The model was trained by optimizing a composite cost function (Multi-Task Loss), which was calculated as the sum of the individual errors: Cross-Entropy (CE) for the multi-class heads and Binary Cross-Entropy (BCE) for the binary heads. The Adam optimizer was used with a learning rate $lr = 10^{-4}$ [4]. Furthermore, an Early Stopping technique was applied with a patience of 4 epochs, monitoring the validation error so as to save the optimal weights of the network [5].

3.3 Performance Evaluation (Test Inference) and Generalization Capability

The final evaluation was performed exclusively on the unseen Test Set. A critical goal of this stage was to evaluate the generalization capability of the model under conditions it had not seen exactly during training. The Test Set included intermediate SNR values (e.g., 13.5 dB or 27.2 dB) and slightly differentiated distortion levels, ensuring that the network learned the true geometric characteristics of the constellations and did not simply "memorize" specific conditions.

Table 1: Overall accuracy of the multi-criteria Convolutional Neural Network (Multi-Task CNN) across the five independent tasks, evaluated on the unseen test set (Test Set).

Task	**Accuracy**
Modulation Type	68.11%
SNR Range/Severity Level	96.88%
Phase Noise Detect	77.35%
I/Q Imbalance Detect	93.23%
Interference Detect	93.59%

Overall, the modulation recognition accuracy reached 68.11%. However, as captured in Figure 4, this performance is inextricably linked to the noise level. While at low SNRs (0–10 dB) the geometric collapse of the constellations limits accuracy, for SNR values above 20 dB, the CNN achieves nearly perfect classification (approaching 96%), proving its ability to distinguish complex schemes (such as 256-QAM and 128-APSK) under realistic conditions. The detailed distribution of classification and channel estimation errors is visualized in the confusion matrices of Figure 5.

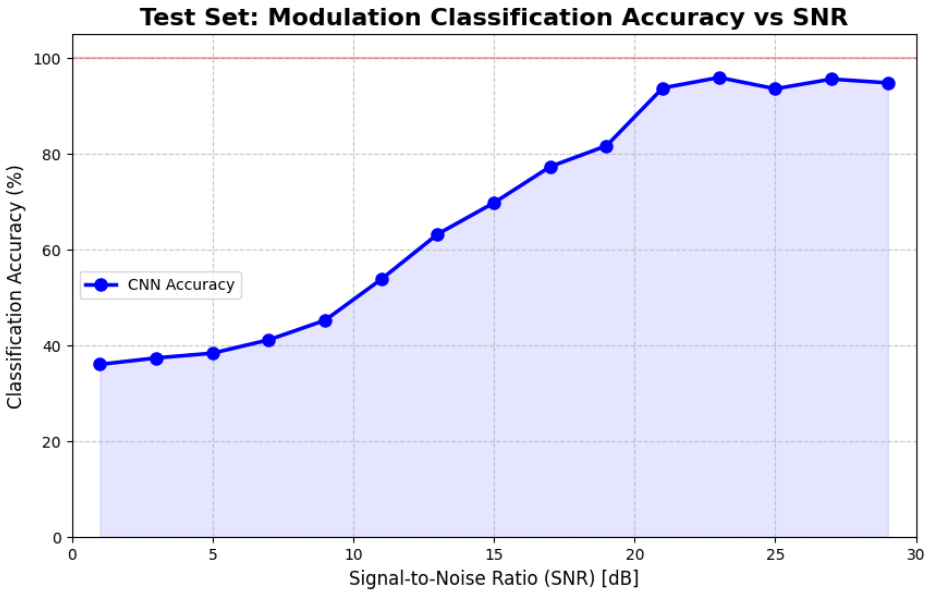


Figure 4: Overall modulation classification accuracy curve (Classification Accuracy) as a function of SNR on the test set. A rapid improvement in network performance is observed for values SNR $>$ 15 dB.

The network demonstrates remarkable robustness at low SNRs ($>$ 15 dB for nearly perfect classification). Simultaneously, the high accuracy in binary classifications (e.g., $>$ 93% in Interference and I/Q Imbalance) proves its sensitivity in detecting even subtle distortions (severity levels) before they completely destroy the constellation.

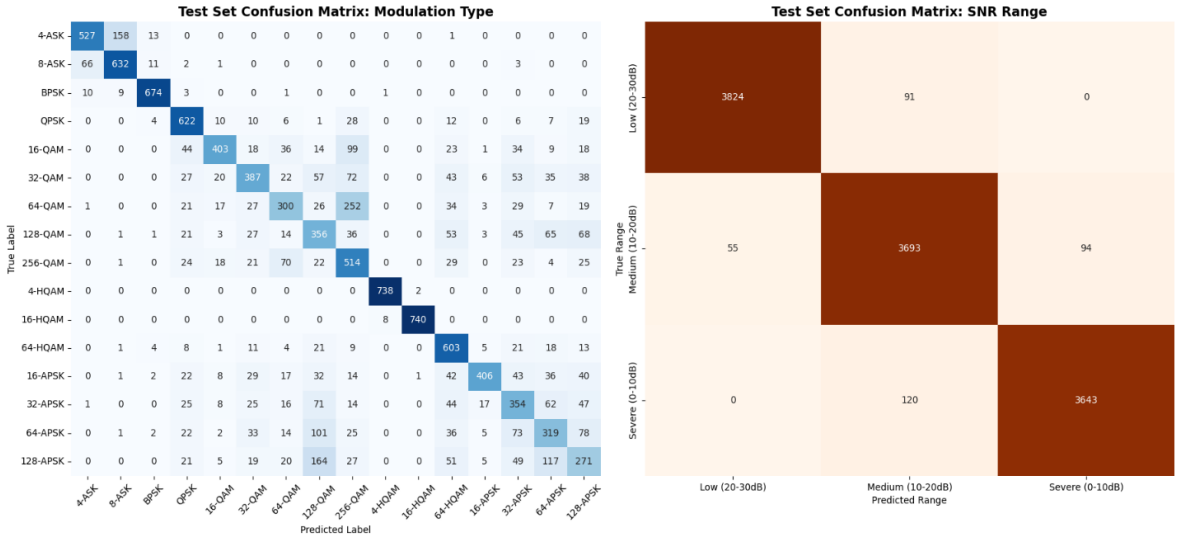


Figure 5: Confusion Matrices from the evaluation on the test set. (Left) Performance in recognizing the modulation type (Modulation Type). (Right) Performance in estimating the channel degradation category (SNR Severity).

3.4 Multimodal Model Integration (Vision-Language Model)

Following modern research trends (RF-GPT [1]), the research was extended to the use of the pretrained `SmolVLM-256M-Instruct`. Due to the high computational requirements of multimodal architectures, random sampling was applied, creating a specialized subset of 6,000 samples

(5,000 images for training and 1,000 for validation). The I/Q histograms were converted into RGB images and Instruction Tuning was applied via a dialogue structure (Chat Template), requesting exclusively the recognition of modulation and three distortions (Phase Noise, I/Q Imbalance, Interference). SNR estimation was deliberately omitted so that the VLM could focus its parametric capacity exclusively on complex geometric deformations. A critical innovation was Dynamic Masking (**Prompt Masking**): via a custom `collate_fn`, the question tokens were assigned the value -100. Thus, they were excluded from the error calculation (Cross-Entropy Loss), "forcing" the network to focus exclusively on generating the correct answer.

3.4.1 LoRA Training & Quantitative Evaluation

Training was implemented via Low-Rank Adaptation (LoRA), comparing two architectures (Ablation Study): the baseline model (**Standard**: $r = 16$, $\alpha = 32$, 3 epochs) and a high-capacity version (**Ultra**: $r = 64$, $\alpha = 128$, 5 epochs, Cosine LR Scheduler), testing the hypothesis that a higher rank r better encodes complex spatial distortions.

Evaluation on 100 unseen samples, with prediction extraction via Regular Expressions (Regex), confirmed the hypothesis (Table 2). The Ultra model improved overall modulation recognition to 64% (+6%). The most impressive improvement was recorded in the detection of I/Q Imbalance (from 78% to 89%), proving that increased capacity "unlocks" a deeper understanding of geometric transformations. A corresponding rise was also noted in Phase Noise detection (+7%).

Table 2: Comparative accuracy evaluation (A/B Testing) between the baseline (Standard, $r = 16$) and the enhanced (Ultra, $r = 64$) multimodal SmolVLM model.

Task	**VLM Standard** ($r = 16$)	**VLM Ultra** ($r = 64$)	**Change**
Modulation Type	58.00%	64.00%	+6.00%
Phase Noise Detect	70.00%	77.00%	+7.00%
I/Q Imbalance Det.	78.00%	89.00%	+11.00%
Interference Detect	90.00%	87.00%	−3.00%

3.4.2 Qualitative Evaluation (Visual Inspection)

The analysis was reinforced through a visual diagnosis algorithm. The Standard network, although capable of identifying multiple distortions (e.g., in 64-QAM, Figure 6), exhibited weaknesses in overlapping transformations, leading to false positive detections (False Positives) of I/Q imbalance in environments of intense Phase Noise and interference (Figure 7).

The Ultra model solved such problems, excellently separating complex distortions (e.g., I/Q Imbalance and Interference in 4-ASK, Figure 8). However, the diagnosis also highlighted the operating limits (edge cases) of the system: the complete spatial collapse of complex constellations (e.g., 128-APSK) due to interference (Figure 9) makes geometric boundaries indistinguishable. This leads the network to unavoidable spatial misinterpretations, such as confusion with 128-QAM and false detection of Phase Noise, underlining the physical limits of optical telecommunication recognition.

SAMPLE #135 (STANDARD)

Metric	Ground Truth	VLM Prediction
Modulation	64QAM	64QAM ✓
Phase Noise	Yes	Yes ✓
I/Q Imbalance	Yes	Yes ✓
Interference	No	No ✓

I/Q Image:

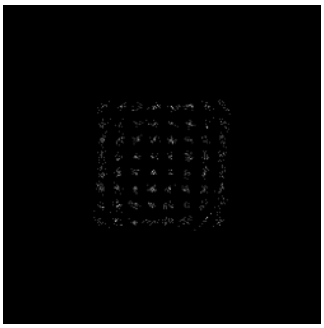


Figure 6: Sample #135 (Standard)

SAMPLE #630 (STANDARD)

Metric	Ground Truth	VLM Prediction
Modulation	4HQAM	4HQAM ✓
Phase Noise	Yes	Yes ✓
I/Q Imbalance	No	Yes ×
Interference	Yes	Yes ✓

I/Q Image:

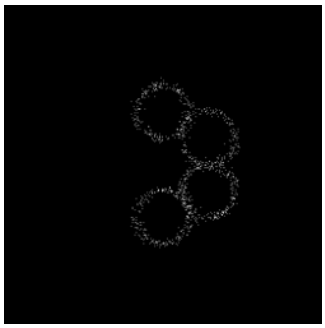


Figure 7: Sample #630 (Standard)

SAMPLE #14 (ULTRA)

Metric	Ground Truth	VLM Prediction
Modulation	4ASK	4ASK ✓
Phase Noise	No	No ✓
I/Q Imbalance	Yes	Yes ✓
Interference	Yes	Yes ✓

I/Q Image:

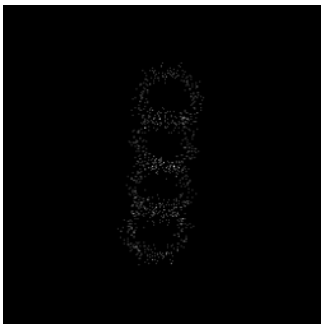


Figure 8: Sample #14 (Ultra)

SAMPLE #759 (ULTRA)

Metric	Ground Truth	VLM Prediction
Modulation	128APSK	128QAM ×
Phase Noise	No	Yes ×
I/Q Imbalance	No	No ✓
Interference	Yes	Yes ✓

I/Q Image:

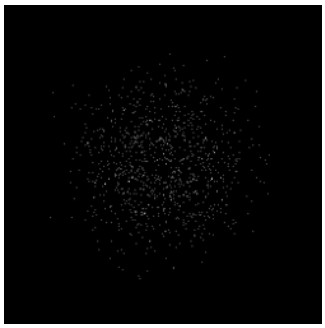


Figure 9: Sample #759 (Ultra)

4. Conclusions: Comparative Analysis of Architectures

The present study evaluated two radically different architectures (a specialized Multi-Task CNN and a generative Multimodal VLM) for the classification and characterization of telecommunication signals. From their comparison, the following critical conclusions are drawn:

i. **Quantitative Superiority (CNN):** The CNN prevailed consistently across all metrics (e.g., 93.2% vs. 89.0% in I/Q Imbalance, 68.1% vs. 64.0% in modulation). This superiority is due to the specialized classification heads, which are optimized purely for statistical prediction, free from the "burden" of text generation (autoregressive decoding) of VLMs.

ii. **Generalization Capability (Sample Efficiency):** Despite lagging behind, the VLM demonstrated impressive efficiency. While the CNN "mapped" the problem by training

on $\sim 53,700$ samples, the VLM achieved accuracies of $\sim 89\%$ having been trained on only $5,000$ images. Achieving these percentages with 10 times fewer data underscores the strong generalization capability of Multimodal Models.

iii. **Computational Complexity:** The VLM is an extremely "heavy" model. As the Ablation study showed, it requires high parametric capacity (the Ultra version prevailed over the simple LoRA $r = 16$) and the use of accelerators (NVIDIA A100), restricting it to offline diagnostic roles. Conversely, the CNN has a minimal memory footprint and nearly zero inference time, making it completely viable for integration into edge devices and real-time analysis.

iv. **Flexibility vs. Accuracy:** The CNN is strictly rigid, requiring radical retraining to incorporate new tasks. Conversely, the VLM is inherently flexible and can be rapidly adapted to new diagnostic requirements via simple language commands (prompting).

5. References

[1]: Zou, H., Tian, Y., Wang, B., Bariah, L., Lasaulce, S., Huang, C., & Debbah, M. (2026). RF-GPT: Teaching AI to See the Wireless World.

[2]: T. K. Oikonomou et al., "CNN-Based Automatic Modulation Classification Under Phase Imperfections," in *IEEE Wireless Communications Letters*, vol. 13, no. 5, pp. 1508–1512, May 2024, doi: 10.1109/LWC.2024.3379198.

[3]: S. Ruder, "An Overview of Multi-Task Learning in Deep Neural Networks," *arXiv preprint arXiv:1706.05098*, 2017.

[4]: D. P. Kingma and J. Ba, "Adam: A Method for Stochastic Optimization," in *3rd International Conference on Learning Representations (ICLR)*, San Diego, CA, USA, 2015.

[5]: I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.